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Semiconductor device structure and method for forming the same

Abstract

A semiconductor device structure is provided. The semiconductor device structure includes a substrate. The semiconductor device structure includes a gate stack formed over the substrate. The semiconductor device structure includes a spacer structure formed over a sidewall of the gate stack. The spacer structure includes a dielectric layer, a silicon rich layer, and a protection layer. The dielectric layer is formed between the gate stack and the silicon rich layer. The silicon rich layer is formed between the dielectric layer and the protection layer. A first atomic percentage of silicon in the silicon rich layer is greater than about 50%. The semiconductor device structure includes a source/drain structure formed over the substrate. The spacer structure is formed between the source/drain structure and the gate stack.

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Background/Summary

PRIORITY CLAIM AND CROSS-REFERENCE (1) This application claims the benefit of U.S. Provisional Application No. 63/257,928, filed on Oct. 20, 2021, the entirety of which is incorporated by reference herein.

BACKGROUND

(1) The semiconductor integrated circuit (IC) industry has experienced rapid growth. Technological advances in IC materials and design have produced generations of ICs. Each generation has smaller and more complex circuits than the previous generation. However, these advances have increased the complexity of processing and manufacturing ICs.

(2) In the course of IC evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometric size (i.e., the smallest component (or line) that can be created using a fabrication process) has decreased. This scaling-down process generally provides benefits by increasing production efficiency and lowering associated costs.

(3) However, since feature sizes continue to decrease, fabrication processes continue to become more difficult to perform. Therefore, it is a challenge to form reliable semiconductor devices at smaller and smaller sizes.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It should be noted that, in accordance with standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIGS. 1A-1E are three-dimensional perspective views of various stages of a process for forming a semiconductor device structure, in accordance with some embodiments.

(3) FIGS. 2A-2J are top views of various stages of a process for forming a semiconductor device structure, in accordance with some embodiments.

(4) FIGS. 2A-1 to 2J-1 are cross-sectional side views illustrating the semiconductor device structure along a sectional line I-I' in FIGS. 2A-2J, in accordance with some embodiments.

(5) FIGS. 2A-2 to 2J-2 are cross-sectional side views illustrating the semiconductor device structure along a sectional line II-II' in FIGS. 2A-2J, in accordance with some embodiments.

(6) FIG. 2B-3 is an enlarged view of a region of the semiconductor device structure of FIG. 2B-1, in accordance with some embodiments.

(7) FIG. 3A is a top view of a semiconductor device structure, in accordance with some embodiments.

(8) FIG. 3B is a cross-sectional side view illustrating the semiconductor device structure along a sectional line 3B-3B' in FIG. 3A, in accordance with some embodiments.

(9) FIGS. 4A-4D are top views of various stages of a process for forming a semiconductor device structure, in accordance with some embodiments.

- (10) FIGS. 4A-1 to 4D-1 are cross-sectional side views illustrating the semiconductor device structure along a sectional line I-I' in FIGS. 4A-4D, in accordance with some embodiments.
- (11) FIGS. 4A-2 to 4D-2 are cross-sectional side views illustrating the semiconductor device structure along a sectional line II-II' in FIGS. 4A-4D, in accordance with some embodiments.
- (12) FIG. 5 is a flow chart of an example method for fabricating an example semiconductor device structure, in accordance with some embodiments.

DETAILED DESCRIPTION

(13) The following disclosure provides many different embodiments, or examples, for implementing different features of the subject matter provided. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(14) Furthermore, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(15) The term “substantially” or “about” in the description, such as in “substantially flat” or in “substantially level with”, etc., will be understood by the person skilled in the art. In some embodiments the adjective substantially may be removed. Where applicable, the term “substantially” may also include embodiments with “entirely”, “completely”, “all”, etc. The term “about” in conjunction with a specific distance or size is to be interpreted so as not to exclude insignificant deviation from the specified distance or size. The term “substantially” or “about” may be varied in different technologies and be in the deviation range understood by the skilled in the art. For example, the term “substantially” or “about” may also relate to 90% of what is specified or higher, such as 95% of what is specified or higher, especially 99% of what is specified or higher, including 100% of what is specified, though the present invention is not limited thereto. Furthermore, terms such as “substantially parallel” or “substantially perpendicular” may be interpreted as not to exclude insignificant deviation from the specified arrangement and may include for example deviations of up to 10°. The word “substantially” does not exclude “completely”, e.g., a composition which is “substantially free” from Y may be completely free from Y.

(16) Some embodiments of the disclosure are described. Additional operations can be provided before, during, and/or after the stages described in these embodiments. Some of the stages that are described can be replaced or eliminated for different embodiments. Additional features can be added to the semiconductor device structure. Some of the features described below can be replaced or eliminated for different embodiments. Although some embodiments are discussed with operations performed in a particular order, these operations may be performed in another logical order.

(17) Embodiments of the disclosure form a semiconductor device structure with FinFETs. The fins may be patterned by any suitable method. For example, the fins may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes.

Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the fins.

(18) FIGS. 1A-1E are three-dimensional perspective views of various stages of a process for forming a semiconductor device structure, in accordance with some embodiments. As shown in FIG. 1A, a substrate **110** is provided, in accordance with some embodiments. The substrate **110** includes, for example, a semiconductor wafer (such as a silicon wafer) or a portion of a semiconductor wafer.

(19) In some embodiments, the substrate **110** is made of an elementary semiconductor material including silicon or germanium in a single crystal, polycrystal, or amorphous structure. In some other embodiments, the substrate **110** is made of a compound semiconductor, such as silicon carbide, gallium arsenide, gallium phosphide, indium phosphide, indium arsenide, an alloy semiconductor, such as SiGe, or GaAsP, or a combination thereof. The substrate **110** may also include multi-layer semiconductors, semiconductor on insulator (SOI) (such as silicon on insulator or germanium on insulator), or a combination thereof.

(20) In some embodiments, the substrate **110** is a device wafer that includes various device elements. In some embodiments, the various device elements are formed in and/or over the substrate **110**. The device elements are not shown in figures for the purpose of simplicity and clarity. Examples of the various device elements include active devices, passive devices, other suitable elements, or a combination thereof. The active devices may include transistors or diodes (not shown) formed at a surface of the substrate **110**. The passive devices include resistors, capacitors, or other suitable passive devices.

(21) For example, the transistors may be metal oxide semiconductor field effect transistors (MOSFET), complementary metal oxide semiconductor (CMOS) transistors, bipolar junction transistors (BJT), high-voltage transistors, high-frequency transistors, p-channel and/or n-channel field effect transistors (PFETs/NFETs), etc. Various processes, such as front-end-of-line (FEOL) semiconductor fabrication processes, are performed to form the various device elements. The FEOL semiconductor fabrication processes may include deposition, etching, implantation, photolithography, annealing, planarization, one or more other applicable processes, or a combination thereof.

(22) In some embodiments, isolation features (not shown) are formed in the substrate **110**. The isolation features are used to define active regions and electrically isolate various device elements formed in and/or over the substrate **110** in the active regions. In some embodiments, the isolation features include shallow trench isolation (STI) features, local oxidation of silicon (LOCOS) features, other suitable isolation features, or a combination thereof.

(23) Afterwards, a first mask layer **122** and a second mask layer **124** may be successively formed over the substrate **110**. In some embodiments, the first mask layer **122** serves a buffer layer or an adhesion layer that is formed between the underlying substrate **110** and the overlying second mask layer **124**. The first mask layer **122** may also be used as an etch stop layer when the second mask layer **124** is removed or etched.

(24) In some embodiments, the first mask layer **122** is made of silicon oxide. In some embodiments, the first mask layer **122** is formed by a deposition process, such as a chemical vapor deposition (CVD) process, a low-pressure chemical vapor deposition (LPCVD) process, a plasma enhanced chemical vapor deposition (PECVD) process, a high-density plasma chemical vapor deposition (HDPCVD) process, a spin-on process, or another applicable process.

(25) In some embodiments, the second mask layer **124** is made of silicon oxide, silicon nitride,

silicon oxynitride, or another applicable material. In some embodiments, the second mask layer **124** is formed by a deposition process, such as a chemical vapor deposition (CVD) process, a low-pressure chemical vapor deposition (LPCVD) process, a plasma enhanced chemical vapor deposition (PECVD) process, a high-density plasma chemical vapor deposition (HDPCVD) process, a spin-on process, or another applicable process.

(26) After the formation of the first mask layer **122** and the second mask layer **124**, the first mask layer **122** and the overlying second mask layer **124** are patterned by a photolithography process and an etching process, so as to expose portions of the substrate **110**. For example, the photolithography process may include photoresist coating (e.g., spin-on coating), soft baking, mask aligning, exposure, post-exposure baking, developing the photoresist, rinsing and drying (e.g., hard baking). Moreover, the etching process may be a dry etching process, such as a reactive ion etching (RIE) process, a neutral beam etching (NBE) process, the like, or a combination thereof.

(27) Afterwards, an etching process is performed on the substrate **110** to remove portions of the substrate **110** by using the patterned first mask layer **122** and the patterned second mask layer **124** as an etch mask, in accordance with some embodiments. After the etching process, the substrate **110** has a base portion **112** and fin portions **114**, in accordance with some embodiments. The fin portions **114** are over the base portion **112**, in accordance with some embodiments. The fin portions **114** are spaced apart from each other, in accordance with some embodiments.

(28) In some embodiments, the etching process includes a dry etching process or a wet etching process. In some embodiments, the substrate **110** is etched by a dry etching process, such as an RIE process, an NBE process, the like, or a combination thereof. The dry etching process may be performed using a process gas including fluorine-based etchant gas. For example, the process gas may include SF₆, C_xF_y (x and y are both integers), NF₃ or a combination thereof.

(29) In some other embodiments (not shown), the fin portions **114** have tapered sidewalls. For example, each of the fin portions **114** has a width that gradually increases from the top portion to the lower portion. The fin portion **114** has opposite sidewalls **114s**, in accordance with some embodiments. The base portion **112** has a top surface **112a**, in accordance with some embodiments.

(30) As shown in FIG. 1A, an isolation layer **130** is formed over the base portion **112**, the fin portions **114**, the first mask layer **122**, and the second mask layer **124**, in accordance with some embodiments. The isolation layer **130** conformally covers the first mask layer **122**, the second mask layer **124**, the top surface **112a** of the base portion **112**, and the sidewalls **114s** of the fin portions **114**, in accordance with some embodiments. The isolation layer **130** has trenches **132**, in accordance with some embodiments.

(31) In some embodiments, the isolation layer **130** is made of oxide (such as silicon oxide), fluorosilicate glass (FSG), a low-k dielectric material, and/or another suitable dielectric material. The isolation layer **130** may be deposited by an atomic layer deposition (ALD) process, a chemical vapor deposition (CVD) process, or another applicable process.

(32) As shown in FIG. 1B, a dielectric layer **140** is formed over the isolation layer **130**, in accordance with some embodiments. The trenches **132** of the isolation layer **130** are filled with the dielectric layer **140**, in accordance with some embodiments.

(33) The dielectric layer **140** is made of oxide (e.g., silicon oxide), nitride (e.g., silicon nitride, silicon carbon nitride, silicon oxycarbon nitride, titanium nitride, or tantalum nitride), carbide (e.g., silicon oxycarbide), metal oxide (e.g., oxides of Li, Be, Mg, Ca, Sr, Sc, Y, Zr, Al, La, Ce, Pr, Nd, Sm, Eu, Gd, Tb, Dy, Ho, Hf, Er, Tm, Yb, Lu, and/or mixtures thereof), or another suitable insulating material, in accordance with some embodiments.

(34) In some embodiments, the isolation layer **130** and the dielectric layer **140** are made of different materials with different etching rates under an etchant. The dielectric layer **140** is formed using a deposition process, such as a chemical vapor deposition (CVD) process, an atomic layer deposition (ALD) process, or a physical vapor deposition (PVD) process, in accordance with some

embodiments.

(35) As shown in FIGS. 1B and 1C, top portions of the isolation layer **130** and the dielectric **140**, the first mask layer **122**, and the second mask layer **124** are removed, in accordance with some embodiments. In some embodiments, top portions of the fin portions **114** are also removed.

(36) After the removal process, the dielectric layer **140** remaining in the trenches **132** forms dielectric fins **142**, in accordance with some embodiments. The dielectric fins **142** are separated from each other by the fin portions **114** and the isolation layer **130**, in accordance with some embodiments. The isolation layer **130** wraps around the dielectric fin **142**, in accordance with some embodiments. The isolation layer **130** separates the dielectric fins **142** from the fin portions **114** and the base portion **112**, in accordance with some embodiments.

(37) In some embodiments, top surfaces **142a**, **114a**, and **134** of the dielectric fins **142**, the fin portions **114**, and the isolation layer **130** are substantially coplanar with (or level with) each other, in accordance with some embodiments. The term “substantially coplanar” in the application may include small deviations from coplanar geometries. The deviations may be due to manufacturing processes.

(38) The removal process includes performing a thinning process on a top surface **141** of the dielectric layer **140**, in accordance with some embodiments. The thinning process includes a chemical mechanical polishing (CMP) process, in accordance with some embodiments.

(39) As shown in FIG. 1D, upper portions of the isolation layer **130** are removed, in accordance with some embodiments. After the removal process, the top surface **134** of the isolation layer **130** is lower than the top surfaces **142a** and **114a** of the dielectric fins **142** and the fin portions **114**, in accordance with some embodiments. The removal process includes an etching process, such as a dry etching process or a wet etching process, in accordance with some embodiments.

(40) FIG. 2A is a top view of the semiconductor device structure of FIG. 1E, in accordance with some embodiments. FIG. 2A-1 is a cross-sectional side view illustrating the semiconductor device structure along a sectional line I-I' in FIG. 2A, in accordance with some embodiments. FIG. 2A-2 is a cross-sectional side view illustrating the semiconductor device structure along a sectional line II-II' in FIG. 2A, in accordance with some embodiments.

(41) As shown in FIGS. 1E, 2A, 2A-1, and 2A-2, gate stacks **G1** and **G2** and mask layers **M1** and **M2** are formed over the fin portions **114**, the isolation layer **130**, and the dielectric fins **142**, in accordance with some embodiments. The gate stacks **G1** and **G2** are spaced apart from each other by a gap **GA**, in accordance with some embodiments.

(42) Each of the gate stacks **G1** and **G2** includes a gate dielectric layer **150** and a gate electrode **160**, in accordance with some embodiments. The gate dielectric layer **150** is formed over the fin portions **114**, the isolation layer **130**, and the dielectric fins **142**, in accordance with some embodiments. The gate electrode **160** is formed over the gate dielectric layer **150**, in accordance with some embodiments.

(43) The gate dielectric layer **150** is made of an insulating material, such as oxide (e.g., silicon oxide), in accordance with some embodiments. The gate electrode **160** is made of a semiconductor material (e.g. polysilicon) or a conductive material (e.g., metal or alloy), in accordance with some embodiments.

(44) The formation of the gate dielectric layer **150** and the gate electrode **160** includes: depositing a gate dielectric material layer (not shown) over the fin portions **114**, the isolation layer **130**, and the dielectric fins **142**; depositing a gate electrode material layer (not shown) over the gate dielectric material layer; sequentially forming the mask layers **M1** and **M2** over the gate electrode material layer, wherein the mask layers **M1** and **M2** expose portions of the gate electrode material layer; and removing the exposed portions of the gate electrode material layer and the gate dielectric material layer thereunder, in accordance with some embodiments.

(45) In some embodiments, the mask layer **M1** serves as a buffer layer or an adhesion layer that is formed between the underlying gate electrode **160** and the overlying mask layer **M2**. The mask

layer M1 may also be used as an etch stop layer when the mask layer M2 is removed or etched. The mask layers M1 and M2 are made of different materials, in accordance with some embodiments.

(46) In some embodiments, the mask layer M1 is made of oxide, such as silicon oxide. In some embodiments, the mask layer M1 is formed by a deposition process, such as a chemical vapor deposition (CVD) process, a low-pressure chemical vapor deposition (LPCVD) process, a plasma enhanced chemical vapor deposition (PECVD) process, a high-density plasma chemical vapor deposition (HDPCVD) process, a spin-on process, or another applicable process.

(47) In some embodiments, the mask layer M2 is made of nitride (e.g., silicon nitride), oxynitride (e.g., silicon oxynitride), or another applicable material. In some embodiments, the mask layer M2 is formed by a deposition process, such as a chemical vapor deposition (CVD) process, a low-pressure chemical vapor deposition (LPCVD) process, a plasma enhanced chemical vapor deposition (PECVD) process, a high-density plasma chemical vapor deposition (HDPCVD) process, a spin-on process, or another applicable process.

(48) After the deposition processes for forming the mask layers M1 and M2, the mask layers M1 and M2 are patterned by a photolithography process and an etching process, so as to expose the portions of the gate electrode material layer.

(49) FIGS. 2A-2J are top views of various stages of a process for forming a semiconductor device structure, in accordance with some embodiments. FIGS. 2A-1 to 2J-1 are cross-sectional side views illustrating the semiconductor device structure along a sectional line I-I' in FIGS. 2A-2J, in accordance with some embodiments. FIGS. 2A-2 to 2J-2 are cross-sectional side views illustrating the semiconductor device structure along a sectional line II-II' in FIGS. 2A-2J, in accordance with some embodiments.

(50) After the step of FIG. 2A, as shown in FIGS. 2B, 2B-1, and 2B-2, a dielectric layer 170 is formed over the gate stacks G1 and G2, the mask layers M1 and M2, and the substrate 110, in accordance with some embodiments. The dielectric layer 170 conformally covers sidewalls S1 and S2 of the gate stacks G1 and G2, the mask layers M1 and M2, and a top surface 116 of the substrate 110, in accordance with some embodiments.

(51) The dielectric layer 170 has bottom portions 172, sidewall portions 174, and upper portions 176, in accordance with some embodiments. The bottom portions 172 are over the substrate 110, in accordance with some embodiments. The sidewall portions 174 cover the mask layer M1 and the sidewalls S1 and S2 of the gate stacks G1 and G2, in accordance with some embodiments. The upper portions 176 cover the mask layer M2, in accordance with some embodiments.

(52) In some embodiments, the dielectric layer 170 is made of an insulating material, such as oxide (e.g., silicon oxycarbonitride (SiCON) or silicon oxide), nitride (e.g., silicon nitride), carbide (e.g., silicon carbide), oxynitride (e.g., silicon oxynitride), or another applicable insulating material. In some embodiments, the insulating material includes a low dielectric constant (low-k) material. The low dielectric constant material has a dielectric constant, which is about 5.

(53) The dielectric layer 170 is formed using a deposition process, such as a chemical vapor deposition (CVD) process, an atomic layer deposition (ALD) process, or a physical vapor deposition (PVD) process, in accordance with some embodiments.

(54) As shown in FIGS. 2B, 2B-1, and 2B-2, a dielectric layer 180 is formed over the dielectric layer 170, in accordance with some embodiments. The dielectric layer 180 conformally covers an upper surface 171 of the dielectric layer 170, in accordance with some embodiments. In some embodiments, the density of the dielectric layer 170 is greater than that of the dielectric layer 180, which improves the anti-etching ability of the dielectric layer 170.

(55) In some embodiments, the dielectric constant of the dielectric layer 180 is lower than that of the dielectric layer 170, and the dielectric layer 180 is thicker than the dielectric layer 170, which reduce the dielectric constant of a spacer structure including the dielectric layers 170 and 180 and therefore reduces RC (referring to a time constant that is a product of resistance and capacitance) delay, in accordance with some embodiments.

(56) FIG. 2B-3 is an enlarged view of a region R of the semiconductor device structure of FIG. 2B-1, in accordance with some embodiments. As shown in FIG. 2B-3, the thickness T170 of the dielectric layer 170 is less than the thickness T180 of the dielectric layer 180, in accordance with some embodiments.

(57) The dielectric layer 180 has bottom portions 182, sidewall portions 184, and upper portions 186, in accordance with some embodiments. The bottom portions 182 are disposed over the bottom portions 172 of the dielectric layer 170, in accordance with some embodiments. The sidewall portions 184 are disposed over the sidewall portions 174 of the dielectric layer 170, in accordance with some embodiments. The upper portions 186 are disposed over the upper portions 176 of the dielectric layer 170, in accordance with some embodiments.

(58) As shown in FIG. 2B-2, since the dielectric layer 180 conformally covers the upper surface 171 of the dielectric layer 170, the dielectric layer 180 has trenches 183, in accordance with some embodiments. Each trench 183 is located between the corresponding dielectric fin 142 and the corresponding fin portion 114, in accordance with some embodiments.

(59) In some embodiments, the dielectric layer 180 is made of an insulating material, such as oxide (e.g., silicon oxycarbonitride (SiCON) or silicon oxide), nitride (e.g., silicon nitride), carbide (e.g., silicon carbide), oxynitride (e.g., silicon oxynitride), or another applicable insulating material. In some embodiments, the insulating material includes a low dielectric constant (low-k) material. The low dielectric constant material has a dielectric constant, which is lower than 5, in accordance with some embodiments.

(60) The dielectric layer 180 is formed using a deposition process, such as a chemical vapor deposition (CVD) process, an atomic layer deposition (ALD) process, or a physical vapor deposition (PVD) process, in accordance with some embodiments.

(61) As shown in FIGS. 2B, 2B-1, and 2B-2, a silicon rich layer 190 is formed over the dielectric layer 180, in accordance with some embodiments. The silicon rich layer 190 is used as an etch stop layer in the subsequent etching process, in accordance with some embodiments. The silicon rich layer 190 conformally covers an upper surface 181 of the dielectric layer 180, in accordance with some embodiments.

(62) The silicon rich layer 190 has bottom portions 192, sidewall portions 194, and upper portions 196, in accordance with some embodiments. The bottom portions 192 are disposed over the bottom portions 182 of the dielectric layer 180, in accordance with some embodiments. The sidewall portions 194 are disposed over the sidewall portions 184 of the dielectric layer 180, in accordance with some embodiments. The upper portions 196 are disposed over the upper portions 186 of the dielectric layer 180, in accordance with some embodiments.

(63) Since the silicon rich layer 190 is used as an etch stop layer in the subsequent etching process, the silicon rich layer 190 is able to prevent the bottom portions 182 of the dielectric layer 180 from being over-etched in the subsequent etching process, which improves the yield of the subsequent etching process, in accordance with some embodiments.

(64) As shown in FIG. 2B-2, since the silicon rich layer 190 is used as an etch stop layer in the subsequent etching process, the silicon rich layer 190 in the trenches 183 is able to prevent the dielectric layer 180 under the trenches 183 from being over-etched in the subsequent etching process, which improves the yield of the subsequent etching process, in accordance with some embodiments.

(65) As shown in FIG. 2B-3, the silicon rich layer 190 has a thickness T190, in accordance with some embodiments. The thickness T190 ranges from about 2.3 Å to about 12 Å, in accordance with some embodiments. The thickness T190 is less than the thickness T180 of the dielectric layer 180, in accordance with some embodiments.

(66) The atomic percentage of silicon in the silicon rich layer 190 is greater than about 50%, in accordance with some embodiments. The atomic percentage of silicon in the silicon rich layer 190 ranges from about 51% to about 100%, in accordance with some embodiments. The atomic

percentage of silicon in the silicon rich layer **190** ranges from about 60% to about 100%, in accordance with some embodiments. The atomic percentage of silicon in the silicon rich layer **190** ranges from about 80% to about 100%, in accordance with some other embodiments.

(67) In some embodiments, as shown in FIG. 2B-1, the atomic percentage of silicon in the silicon rich layer **190** decreases from a lower surface **191b** to an upper surface **191** of the silicon rich layer **190**. The atomic percentage of silicon in the silicon rich layer **190** close to the lower surface **191b** is about 100%, in accordance with some embodiments.

(68) When the atomic percentage of silicon in the silicon rich layer **190** is about 100%, the silicon rich layer **190** is also referred to as a semiconductor layer or a silicon layer, in accordance with some embodiments. In some embodiments, the silicon rich layer **190** is made of silicon.

(69) In some other embodiments, the silicon rich layer **190** is made of a silicon rich material, such as silicon rich nitride. The atomic percentage of silicon in silicon rich nitride is greater than the atomic percentage of nitrogen in silicon rich nitride, in accordance with some embodiments.

(70) The silicon rich layer **190** is formed using a deposition process, such as an atomic layer deposition process, a chemical vapor deposition process, or a physical vapor deposition process, in accordance with some embodiments.

(71) In some embodiments, the silicon rich layer **190** is made of silicon and formed using the atomic layer deposition process, the dielectric layer **180** is soaked in a silicon containing precursor gas, such as silane or SiH_2Cl_2 , during the atomic layer deposition process.

(72) In some embodiments, the silicon rich layer **190** is made of silicon rich nitride and formed using the atomic layer deposition process, the upper surface **181** of the dielectric layer **180** is alternatively reacted with a silicon-containing precursor gas and a nitrogen-containing precursor gas in the same process chamber. The nitrogen containing precursor gas includes ammonia (NH_3), in accordance with some embodiments.

(73) During the atomic layer deposition process, the number of times the upper surface **181** is reacted with the silicon-containing precursor gas is greater than the number of times the upper surface **181** is reacted with the nitrogen-containing precursor gas, in accordance with some embodiments. Therefore, the silicon rich layer **190** has a first number of monoatomic silicon layers and a second number of monoatomic nitrogen layers, and the first number is greater than the second number, in accordance with some embodiments.

(74) As shown in FIGS. 2B, 2B-1, and 2B-2, a protection layer **210** is formed over the silicon rich layer **190**, in accordance with some embodiments. The protection layer **210** conformally covers the upper surface **191** of the silicon rich layer **190**, in accordance with some embodiments.

(75) The silicon rich layer **190** is disposed between the dielectric layer **180** and the protection layer **210**, in accordance with some embodiments. The silicon rich layer **190** separates the dielectric layer **180** from the protection layer **210**, in accordance with some embodiments.

(76) The protection layer **210** has bottom portions **212**, sidewall portions **214**, and upper portions **216**, in accordance with some embodiments. The bottom portions **212** are disposed over the bottom portions **192** of the silicon rich layer **190**, in accordance with some embodiments. The sidewall portions **214** are disposed over the sidewall portions **194** of the silicon rich layer **190**, in accordance with some embodiments. The upper portions **216** are disposed over the upper portions **196** of the silicon rich layer **190**, in accordance with some embodiments.

(77) As shown in FIGS. 2B-1 and 2B-3, the thickness **T210** of the protection layer **210** is greater than the thickness **T190** of the silicon rich layer **190**, in accordance with some embodiments. In some embodiments, a ratio of the thickness **T210** to the thickness **T190** ranges from about 4:1 to about 12:1.

(78) As shown in FIGS. 2B-1 and 2B-3, the thicknesses **T210**, **T190**, **T180**, and **T170** are measured in a direction **A1**, in accordance with some embodiments. The direction **A1** is perpendicular to the sidewall **S2** of the gate stack **G2** or the sidewall **S1** of the gate stack **G1**, in accordance with some embodiments.

(79) The protection layer **210** and the silicon rich layer **190** are made of different materials with different compositions, which improves the etching selectivity of the protection layer **210** to the silicon rich layer **190**, in accordance with some embodiments. The etching selectivity of the protection layer **210** to the silicon rich layer **190** ranges from about 4 to about 7, in accordance with some embodiments.

(80) The protection layer **210** is made of a dielectric material, such as nitride (e.g., silicon nitride), carbide (e.g., silicon carbide), or oxynitride (e.g., silicon oxynitride), in accordance with some embodiments. The protection layer **210** is formed using a deposition process, such as an atomic layer deposition process, a chemical vapor deposition process, or a physical vapor deposition process, in accordance with some embodiments.

(81) In some embodiments, the protection layer **210** and the silicon rich layer **190** are formed in the same chamber, such as an atomic layer deposition chamber. That is, the protection layer **210** and the silicon rich layer **190** are formed in situ, in accordance with some embodiments. The protection layer **210** and the silicon rich layer **190** are formed using atomic layer deposition processes, in accordance with some embodiments.

(82) In some embodiments, the protection layer **210** is made of silicon nitride (Si.sub.3N.sub.4), and the atomic percentage of silicon in silicon nitride is close to 43%. In some embodiments, the atomic percentage of silicon in the silicon rich layer **190** close to the upper surface **191** (or the protection layer **210**) is about 43%.

(83) In some embodiments, the atomic percentage of silicon in the silicon rich layer **190** decreases toward the protection layer **210** and becomes similar to the atomic percentage of silicon in the protection layer **210**. Therefore, the lattice structure of the silicon rich layer **190** close to the protection layer **210** is similar to the lattice structure of the protection layer **210**, which improves the adhesion between the silicon rich layer **190** and the protection layer **210**, in accordance with some embodiments.

(84) As shown in FIGS. 2C, 2C-1, and 2C-2, the top portions **216** and the bottom portions **212** of the protection layer **210** are partially removed, in accordance with some embodiments. After the removal process, the protection layer **210** covers the sidewall portions **194** and exposes the bottom portion **192** and the upper portions **196** of the silicon rich layer **190**, in accordance with some embodiments. The sidewall portions **214** of the protection layer **210** are spaced apart from each other, in accordance with some embodiments. The silicon rich layer **190** is connected between the sidewall portions **214**, in accordance with some embodiments.

(85) As shown in FIGS. 2C, 2C-1, and 2C-2, after the removal process, an oxide layer **220** is formed over the exposed portions of the silicon rich layer **190** exposed by the protection layer **210**, in accordance with some embodiments. The oxide layer **220** is in direct contact with the silicon rich layer **190**, in accordance with some embodiments.

(86) The oxide layer **220** has portions **222** and **223**, in accordance with some embodiments. The portion **222** is located between the sidewall portions **214** of the protection layer **210**, in accordance with some embodiments. The portion **222** is connected to the sidewall portions **214** of the protection layer **210**, in accordance with some embodiments.

(87) The portions **222** cover the bottom portions **192** of the silicon rich layer **190**, in accordance with some embodiments. The portion **222** is partially embedded in the silicon rich layer **190**, in accordance with some embodiments. The portion **222** has an upper surface **222a** and a lower surface **222b**, in accordance with some embodiments. The upper surface **222a** is higher than the upper surface **192a** of the bottom portion **192** of the silicon rich layer **190**, in accordance with some embodiments. The lower surface **222b** is lower than the upper surface **192a**, in accordance with some embodiments.

(88) The bottom portion **192** of the silicon rich layer **190** is partially between the portion **222** of the oxide layer **220** and the bottom portion **182** of the dielectric layer **180**, in accordance with some embodiments. The thickness **T220** of the oxide layer **220** ranges from about 4 Å to about 12 Å, in

accordance with some embodiments. The portions **223** cover top surfaces **196a** of the upper portions **196** of the silicon rich layer **190**, in accordance with some embodiments.

(89) The oxide layer **220** is formed by oxidation of the surface part of the silicon rich layer **190** exposed by the protection layer **210**, and therefore the oxide layer **220** and the silicon rich layer **190** have the same semiconductor element, in accordance with some embodiments. The semiconductor element includes silicon, in accordance with some embodiments. The oxide layer **220** is made of silicon oxide or silicon oxide nitride, in accordance with some embodiments.

(90) The removal process includes an etching process, in accordance with some embodiments. The etching process includes an anisotropic etching process, such as a dry etching process (e.g., a plasma etching process), in accordance with some embodiments.

(91) The etching process uses an etching gas including oxygen, carbon, hydrogen, and fluorine, in accordance with some embodiments. When the protection layer **210** is etched through and therefore exposes the silicon rich layer **190** thereunder, the exposed surface part of the silicon rich layer **190** is oxidized to form the oxide layer **220**, in accordance with some embodiments. Since volume expansion due to the oxidation occurs, the volume of the oxide layer **220** is greater than the volume of the oxidized surface part, in accordance with some embodiments.

(92) In some embodiments, an etching rate of the oxide layer **220** is lower than an etching rate of the protection layer **210** in the etching process. The etching selectivity of silicon nitride to silicon oxide (or silicon oxide nitride) is about 4 to about 7, in accordance with some embodiments.

(93) Since the etching selectivity of silicon nitride to silicon oxide (or silicon oxide nitride) is high, the etching process is stopped at the oxide layer **220**, in accordance with some embodiments. Therefore, the formation of the oxide layer **220** prevents the corner portions **170c** and **180c** of the dielectric layers **170** and **180** from being etched, in accordance with some embodiments. Therefore, the yield of the etching process is improved, in accordance with some embodiments. The oxide layer **220** is also referred to as a buffer layer, in accordance with some embodiments.

(94) As shown in FIGS. 2D, 2D-1, and 2D-2, the oxide layer **220** and the top portions **196**, **186** and **176** and the bottom portions **192**, **182**, and **172** of the silicon rich layer **190**, the dielectric layer **180**, and the dielectric layer **170** are partially removed, in accordance with some embodiments.

(95) As shown in FIG. 2D-2, after the removal process, the remaining silicon rich layer **190** is embedded in the remaining dielectric layer **180**, in accordance with some embodiments. The remaining protection layer **210** is embedded in the remaining silicon rich layer **190**, in accordance with some embodiments.

(96) The removal process includes an etching process, in accordance with some embodiments. The etching process includes an anisotropic etching process, such as a dry etching process (e.g., a plasma etching process), in accordance with some embodiments.

(97) As shown in FIGS. 2D, 2D-1, and 2D-2, an oxide layer **220'** is formed over the exposed portions of the silicon rich layer **190** exposed by the protection layer **210**, in accordance with some embodiments. The oxide layer **220'** is in direct contact with the silicon rich layer **190**, in accordance with some embodiments.

(98) The oxide layer **220'** has portions **224**, **226** and **228**, in accordance with some embodiments. The portions **224** cover top surfaces **194a** of the sidewall portions **194** of the silicon rich layer **190**, in accordance with some embodiments. The portion **224** has a sloped upper surface **224a**, in accordance with some embodiments.

(99) The sloped upper surface **224a** is a convex curved surface, in accordance with some embodiments. The sloped upper surface **224a** protrudes from the upper surfaces **184a** of the sidewall portions **184** and **214** of the dielectric layer **180** and the protection layer **210**, in accordance with some embodiments.

(100) The portions **226** of the oxide layer **220'** cover sidewalls **192b** of the bottom portions **192** of the silicon rich layer **190**, in accordance with some embodiments. The portion **226** has a sidewall **226a**, in accordance with some embodiments. The sidewall **226a** is substantially vertical to the top

surface **114a** of the fin portion **114**, in accordance with some embodiments.

(101) The sidewall **226a** is a convex curved surface, in accordance with some embodiments. The sidewall **226a** protrudes from the sidewalls **182a** and **214a** of the bottom portions **182** and the sidewall portions **214** of the dielectric layer **180** and the protection layer **210**, in accordance with some embodiments.

(102) As shown in FIG. 2D-2, the portions **228** of the oxide layer **220'** cover the upper surface **191** of the silicon rich layer **190** between the dielectric fins **142** and the fin portions **114**, in accordance with some embodiments. The portion **228** has an upper surface **228a**, in accordance with some embodiments. The upper surface **228a** is a convex curved surface, in accordance with some embodiments. The upper surface **228a** protrudes from the upper surfaces **181** and **211** of the dielectric layer **180** and the protection layer **210**, in accordance with some embodiments.

(103) The remaining protection layer **210**, the remaining silicon rich layer **190**, the remaining dielectric layer **180**, the remaining dielectric layer **170**, and the oxide layer **220'** together form a spacer structure S, in accordance with some embodiments. The spacer structure S remains over the sidewalls S1 and S2 of the gate stacks G1 and G2 and opposite sidewalls of the mask layers M1 and M2, in accordance with some embodiments. In some embodiments, the spacer structure S surrounds the gate stacks G1 and G2.

(104) As shown in FIG. 2D-2, a height H1 of the spacer structure S is less than a height H2 of the dielectric fin **142** relative to the top surface **134** of the isolation layer **130**, in accordance with some embodiments. In some embodiments, a ratio of the height H1 to the height H2 ranges from about 0.2:1 to about 0.4:1.

(105) Afterwards, as shown in FIGS. 2E, 2E-1, and 2E-2, the fin portions **114** exposed by the gate stacks G1 and G2 and the spacer structure S are partially removed to form recesses **114r** in the fin portions **114**, in accordance with some embodiments. The removal process includes an etching process, such as a dry etching process or a wet etching process, in accordance with some embodiments.

(106) As shown in FIGS. 2F, 2F-1, and 2F-2, source/drain structures SD (which may be referred to as a region that may be a source region or a drain region in some embodiments) are respectively formed in the recesses **114r** of the fin portions **114**, in accordance with some embodiments. The portions **226** of the oxide layer **220'** are in direct contact with the source/drain structures SD, in accordance with some embodiments. In some embodiments, as shown in FIG. 2F-2, the spacer structure S is over sidewalls SD2 of the source/drain structures SD, in accordance with some embodiments. In some embodiments, as shown in FIG. 2F-2, voids V are formed.

(107) Each void V is surrounded by the corresponding source/drain structure SD, the spacer structure S, and the corresponding dielectric fin **142**, in accordance with some embodiments. The voids V are substantially closed, in accordance with some embodiments. The source/drain structure SD is located between two adjacent voids V, in accordance with some embodiments.

(108) In some embodiments, the source/drain structures SD are made of a semiconductor material (e.g., silicon germanium), in accordance with some embodiments. The source/drain structures SD may have P-type dopants, such as the Group IIIA element. The Group IIIA element includes boron or another suitable material.

(109) In some other embodiments, the source/drain structures SD are made of a semiconductor material (e.g., silicon), in accordance with some embodiments. The source/drain structures SD may have N-type dopants, such as the Group VA element. The Group VA element includes phosphor (P), antimony (Sb), or another suitable Group VA material. The source/drain structures SD are formed using an epitaxial process, in accordance with some embodiments.

(110) As shown in FIG. 2F-2, the spacer structure S is located between the corresponding source/drain structure SD and the corresponding dielectric fin **142**, in accordance with some embodiments. The source/drain structure SD is located between two adjacent dielectric fins **142**, in accordance with some embodiments. The source/drain structure SD is in direct contact with the two

adjacent dielectric fins **142**, in accordance with some embodiments. The dielectric fins **142** are used to limit the maximum width of the source/drain structure SD therebetween, in accordance with some embodiments.

(111) As shown in FIGS. **2G**, **2G-1**, and **2G-2**, the portions **224** of the oxide layer **220'** and upper portions of the silicon rich layer **190** and the protection layer **210** outside of the trenches TR between the source/drain structures SD and the dielectric layer **180** are removed, in accordance with some embodiments.

(112) The removal process is able to reduce the aspect ratio of the gap GA, in accordance with some embodiments. Since the dielectric constants of the silicon rich layer **190** and the protection layer **210** are greater than that of the dielectric layers **170** and **180**, the removal process is able to reduce the overall dielectric constant of the spacer structure S and therefore reduces RC delay, in accordance with some embodiments.

(113) After the removal process, lower portions of the silicon rich layer **190** and the protection layer **210** remain in the trenches TR between the source/drain structures SD and the dielectric layer **180**, in accordance with some embodiments. The upper surface **211** of the protection layer **210** is substantially level with the upper surface SD1 of the source/drain structure SD, in accordance with some embodiments.

(114) As shown in FIGS. **2G** and **2G-1**, an oxide layer **229** is formed over the silicon rich layer **190**, in accordance with some embodiments. The oxide layer **229** is formed by oxidation of the exposed surface part of the silicon rich layer **190**, and therefore the oxide layer **229** and the silicon rich layer **190** have the same semiconductor element, in accordance with some embodiments.

(115) The semiconductor element includes silicon, in accordance with some embodiments. The oxide layer **229** is made of silicon oxide or silicon oxide nitride, in accordance with some embodiments. The removal process includes an etching process such as a wet etching process or a dry etching process, in accordance with some embodiments.

(116) As shown in FIGS. **2G-1** and **2G-2**, an etch stop layer **230** is conformally formed over the source/drain structure SD, the dielectric fins **142**, the spacer structure S, and the mask layer M2, in accordance with some embodiments. The etch stop layer **230** is made of an insulating material, such as a nitrogen-containing material (e.g., silicon nitride), in accordance with some embodiments.

(117) The etch stop layer **230** is formed using a deposition process, such as a chemical vapor deposition (CVD) process, an atomic layer deposition (ALD) process, or a physical vapor deposition (PVD) process, in accordance with some embodiments.

(118) As shown in FIGS. **2G**, **2G-1**, and **2G-2**, a dielectric layer **240** is formed over the etch stop layer **230**, in accordance with some embodiments. The dielectric layer **240** is made of any suitable insulating material, such as silicon oxide, silicon oxynitride, borosilicate glass (BSG), phosphoric silicate glass (PSG), borophosphosilicate glass (BPSG), fluorinated silicate glass (FSG), low-k material, porous dielectric material, or a combination thereof. The dielectric layer **240** is deposited by any suitable process, such as a CVD process, a spin-on process, a sputtering process, or a combination thereof, in accordance with some embodiments.

(119) Afterwards, as shown in FIGS. **2H**, **2H-1**, and **2H-2**, the mask layers M1 and M2 and upper portions of the dielectric layer **240**, the etch stop layer **230**, the spacer structure S, and the gate stacks G1 and G2 are removed, in accordance with some embodiments.

(120) After the removal process, top surfaces A1, A2, **178**, **188**, **232**, and **242** of the gate stacks G1 and G2, the dielectric layer **170**, the dielectric layer **180**, the etch stop layer **230**, and the dielectric layer **240** are substantially level with each other or substantially coplanar with each other, in accordance with some embodiments.

(121) Thereafter, a gate replacement process is performed, in accordance with some embodiments. As shown in FIGS. **2H**, **2I**, and **2I-1**, the gate stacks G1 and G2 are removed, in accordance with some embodiments. The spacer structure S has trenches **179**, in accordance with some

embodiments. The trenches **179** expose the dielectric fins **142** and the fin portions **114**, and in accordance with some embodiments.

(122) As shown in FIGS. **2I**, **2I-1**, and **2I-2**, gate stacks **G3** and **G4** are formed in the trenches **179**, in accordance with some embodiments. In some embodiments, a ratio of the width W_{subSD} of the source/drain structure **SD** to the width W_{subG} of the gate stack **G3** or **G4** ranges from about 1 to about 2. Each of the gate stacks **G3** and **G4** includes a gate dielectric layer **250**, a work function metal layer **260**, and a gate electrode layer **270**, in accordance with some embodiments. The gate dielectric layer **250** is conformally formed in the trenches **179**, in accordance with some embodiments.

(123) The gate dielectric layer **250** is made of a dielectric material, such as a high dielectric constant (high-k) material, in accordance with some embodiments. The high-k material is made of metal oxides, such as hafnium oxide (HfO_{2}), hafnium silicon oxide ($HfSiO$), hafnium silicon oxynitride ($HfSiON$), hafnium tantalum oxide ($HfTaO$), hafnium titanium oxide ($HfTiO$), hafnium zirconium oxide ($HfZrO$), aluminum oxide, hafnium dioxide-alumina (HfO_{2} — $Al_{2}O_{3}$) alloy, other suitable high-k dielectric materials, or combinations thereof, in accordance with some embodiments. In some other embodiments, the high-k material is made of metal nitrides, metal silicates, transition metal-oxides, transition metal-nitrides, transition metal-silicates, oxynitrides of metals, other suitable materials, or combinations thereof.

(124) The work function metal layer **260** is conformally formed over the gate dielectric layer **250**, in accordance with some embodiments. The work function metal layer **260** provides a desired work function for transistors to enhance device performance including improved threshold voltage. In the embodiments of forming an NMOS transistor, the work function metal layer **260** can be an n-type metal capable of providing a work function value suitable for the device, such as equal to or less than about 4.5 eV. The n-type metal may be made of metal, metal carbide, metal nitride, or a combination thereof. For example, the n-type metal is made of tantalum, tantalum nitride, or a combination thereof.

(125) On the other hand, in the embodiments of forming a PMOS transistor, the work function metal layer **260** can be a p-type metal capable of providing a work function value suitable for the device, such as equal to or greater than about 4.8 eV. The p-type metal may be made of metal, metal carbide, metal nitride, other suitable materials, or a combination thereof. For example, the p-type metal is made of titanium, titanium nitride, other suitable materials, or combinations thereof. The work function metal layer **260** may also be made of hafnium, zirconium, titanium, tantalum, aluminum, metal carbides (e.g., hafnium carbide or zirconium carbide), aluminides, ruthenium or a combination thereof.

(126) The gate electrode layer **270** is formed over the work function metal layer **260**, in accordance with some embodiments. The gate electrode layer **270** is also called a metal gate electrode layer, in accordance with some embodiments. The gate electrode layer **270** is made of a suitable metal material, such as aluminum, tungsten, gold, platinum, cobalt, other suitable metal, an alloy thereof, or a combination thereof, in accordance with some embodiments.

(127) Thereafter, as shown in FIGS. **2J**, **2J-1**, and **2J-2**, portions of the dielectric layer **240** and the etch stop layer **230** are removed to form contact holes **H** in the dielectric layer **240** and the etch stop layer **230**, in accordance with some embodiments. The contact holes **H** pass through the dielectric layer **240** and the etch stop layer **230**, in accordance with some embodiments. The contact hole **H** partially exposes the upper surface **SD1** of the source/drain structure **SD** thereunder, in accordance with some embodiments.

(128) Afterwards, a metal layer (not shown) is formed over the source/drain structure **SD**, in accordance with some embodiments. The metal layer is made of Ti, Co, Ru, or another suitable metal material. The metal layer is formed using a deposition process, such as a physical vapor deposition process, a plating process, another suitable method, or a combination thereof, in accordance with some embodiments.

(129) Thereafter, as shown in FIGS. 2J-1 and 2J-2, the metal layer and the source/drain structures SD are annealed to react the metal layer with the source/drain structures SD so as to form a silicide layer **280** between the metal layer and the source/drain structures SD, in accordance with some embodiments. The silicide layer **280** is in direct contact with the source/drain structures SD, in accordance with some embodiments. The silicide layer **280** includes $\text{TiSi}_{1.2}$ (titanium disilicide), $\text{CoSi}_{1.2}$, or RuSi , in accordance with some embodiments.

(130) As shown in FIGS. 2J-1 and 2J-2, portions of the metal layer that have not reacted with the source/drain structures SD are removed, in accordance with some embodiments. The removal process includes an etching process such as a wet etching process or a dry etching process, in accordance with some embodiments.

(131) As shown in FIGS. 2J, 2J-1, and 2J-2, contact structures **290** are formed in the contact holes H, in accordance with some embodiments. The contact structures **290** pass through the dielectric layer **240** and the etch stop layer **230** to connect to the silicide layer **280** thereunder, in accordance with some embodiments.

(132) The formation of the contact structures **290** includes depositing a conductive material layer (not shown) over the dielectric layer **240** and in the contact holes H; and performing a chemical mechanical polishing (CMP) process over the conductive material layer to remove the conductive material layer outside of the contact holes H, in accordance with some embodiments.

(133) The contact structures **290** are made of tungsten (W) or another suitable conductive material, in accordance with some embodiments. In this step, a semiconductor device structure **200** is substantially formed, in accordance with some embodiments.

(134) FIG. 3A is a top view of a semiconductor device structure **300**, in accordance with some embodiments. FIG. 3B is a cross-sectional side view illustrating the semiconductor device structure **300** along a sectional line 3B-3B' in FIGS. 3A, in accordance with some embodiments.

(135) As shown in FIGS. 3A and 3B, the semiconductor device structure **300** is similar to the semiconductor device structure **200** of FIG. 2J-2, except that the spacer structure S of the semiconductor device structure **300** has recesses R1 between the dielectric fins **142** and the source/drain structures SD, in accordance with some embodiments.

(136) In some embodiments, the protection layer **210** is made of nitride, and the dielectric layers **170** and **180** are made of oxide. Since the etching rate of nitride is greater than the etching rate of oxide in the etching process, the recesses R1 may be formed in the protection layer **210** after the etching process of FIG. 2D-2.

(137) FIGS. 4A-4D are top views of various stages of a process for forming a semiconductor device structure, in accordance with some embodiments. FIGS. 4A-1 to 4D-1 are cross-sectional side views illustrating the semiconductor device structure along a sectional line I-I' in FIGS. 4A-4D, in accordance with some embodiments. FIGS. 4A-2 to 4D-2 are cross-sectional side views illustrating the semiconductor device structure along a sectional line II-II' in FIGS. 4A-4D, in accordance with some embodiments.

(138) After the step of FIG. 2B, as shown in FIGS. 4A, 4A-1 and 4A-2, a silicon rich layer **410** is formed over the protection layer **210**, in accordance with some embodiments. The silicon rich layer **410** conformally covers the upper surface **211** of the protection layer **210**, in accordance with some embodiments. The silicon rich layer **410** has bottom portions **412**, sidewall portions **414**, and upper portions **416**, in accordance with some embodiments.

(139) The bottom portions **412** are over the bottom portions **212** of the protection layer **210**, in accordance with some embodiments. The sidewall portions **414** are over the sidewall portions **214** of the protection layer **210**, in accordance with some embodiments. The upper portions **416** are over the upper portions **216** of the protection layer **210**, in accordance with some embodiments. As shown in FIG. 4A-1, the silicon rich layer **410** has a thickness T_{410} ranging from about 2.3 Å to about 12 Å, in accordance with some embodiments.

(140) The material and the forming method of the silicon rich layer **410** are similar to or the same

as that of the silicon rich layer **190**, in accordance with some embodiments. In some embodiments, the protection layer **210** and the silicon rich layers **190** and **410** are formed in the same chamber, such as an atomic layer deposition chamber. That is, the protection layer **210** and the silicon rich layers **190** and **410** are formed in situ, in accordance with some embodiments. The protection layer **210** and the silicon rich layers **190** and **410** are formed using atomic layer deposition processes, in accordance with some embodiments.

(141) As shown in FIG. 4A-2, the protection layer **210** conformally covers the upper surface **191** of the silicon rich layer **190** and has trenches **213** in the trenches **193** of the silicon rich layer **190**, in accordance with some embodiments. The silicon rich layer **410** is formed over the upper surface **211** of the protection layer **210** and in the trenches **213**, in accordance with some embodiments.

(142) In some embodiments, as shown in FIG. 4A-2, if the trench **193** (e.g. the left one) is narrower or the protection layer **210** is thicker, the trench **193** is filled with the protection layer **210** and therefore the trench **213** is not formed.

(143) Afterwards, as shown in FIGS. 4B, 4B-1 and 4B-2, the step of FIG. 2C is performed to partially remove the top portions **216** and **416** and the bottom portions **212** and **412** of the protection layer **210** and the silicon rich layer **410**, in accordance with some embodiments.

(144) After the removal process, the protection layer **210** and the silicon rich layer **410** cover the sidewall portions **194** and expose the bottom portions **192** and the upper portions **196** of the silicon rich layer **190**, in accordance with some embodiments.

(145) As shown in FIGS. 4B, 4B-1, and 4B-2, after the removal process, an oxide layer **220** is formed over the exposed portions of the silicon rich layer **190**, and an oxide layer **420** is formed over the silicon rich layer **410**, in accordance with some embodiments.

(146) The oxide layer **220** is in direct contact with the silicon rich layer **190**, in accordance with some embodiments. The oxide layer **420** is in direct contact with the silicon rich layer **410**, in accordance with some embodiments.

(147) The oxide layer **220** has portions **222** and **223**, in accordance with some embodiments. The portions **222** are formed over the bottom portions **192** of the silicon rich layer **190**, in accordance with some embodiments. The portions **223** are formed over the upper portions **196** of the silicon rich layer **190**, in accordance with some embodiments.

(148) The oxide layer **420** is formed by oxidation of the exposed surface part of the silicon rich layer **410**, and therefore the oxide layer **420** and the silicon rich layer **410** have the same semiconductor element, in accordance with some embodiments. The semiconductor element includes silicon, in accordance with some embodiments. The oxide layer **420** is made of silicon oxide or silicon oxide nitride, in accordance with some embodiments.

(149) Since the etching selectivity of silicon nitride to silicon oxide (or silicon oxide nitride) is high, the etching process is stopped at the oxide layers **220** and **420**, in accordance with some embodiments. Therefore, the formation of the oxide layers **220** and **420** prevents over etching of the protection layer **210**, which protects the corner portions **170c** and **180c** of the dielectric layers **170** and **180** from being etched, in accordance with some embodiments. Therefore, the yield of the etching process is improved, in accordance with some embodiments.

(150) Furthermore, the formation of the silicon rich layer **410** and the oxide layer **420** is able to prevent over etching of the protection layer **210**, which reduces the required thickness of the protection layer **210**, in accordance with some embodiments. Therefore, the aspect ratio of the gap GA between the oxide layer **420** over the gate stack G1 and the oxide layer **420** over the gate stack G2 may be reduced.

(151) As shown in FIGS. 4C, 4C-1, and 4C-2, the step of FIG. 2D is performed to partially remove the oxide layer **220** and the top portions **196**, **186** and **176** and the bottom portions **192**, **182**, and **172** of the silicon rich layer **190**, the dielectric layer **180**, and the dielectric layer **170**, in accordance with some embodiments. In some embodiments (not shown), the oxide layer **420** and the silicon rich layer **410** are partially or completely removed in this step.

(152) As shown in FIGS. 4C, 4C-1, and 4C-2, an oxide layer **220'** is formed over the exposed portions of the silicon rich layer **190** exposed by the protection layer **210**, in accordance with some embodiments. The oxide layer **220'** is in direct contact with the silicon rich layer **190**, in accordance with some embodiments.

(153) The oxide layer **220'** has portions **224**, **226** and **228**, in accordance with some embodiments. The portions **224** cover the top surfaces **194a** of the sidewall portions **194** of the silicon rich layer **190**, in accordance with some embodiments. The portions **226** cover sidewalls **192b** of the bottom portions **192** of the silicon rich layer **190**, in accordance with some embodiments.

(154) As shown in FIG. 4C-2, the portions **228** cover the upper surface **191** of the silicon rich layer **190** between the dielectric fins **142** and the fin portions **114**, in accordance with some embodiments. As shown in FIG. 4C-2, the oxide layer **420** is formed over the silicon rich layer **410** and has an upper surface **422**, in accordance with some embodiments. The upper surface **422** is a convex curved surface, in accordance with some embodiments. The upper surface **422** protrudes from the upper surfaces **181** and **211** of the dielectric layer **180** and the protection layer **210**, in accordance with some embodiments.

(155) As shown in FIGS. 4C-1 and 4C-2, the remaining protection layer **210**, the remaining silicon rich layers **190** and **410**, the remaining dielectric layers **170** and **180**, and the oxide layers **220'** and **420** together form a spacer structure **S**, in accordance with some embodiments. The spacer structure **S** remains over the sidewalls **S1** and **S2** of the gate stacks **G1** and **G2** and opposite sidewalls of the mask layers **M1** and **M2**, in accordance with some embodiments.

(156) As shown in FIGS. 4D, 4D-1 and 4D-2, the steps of FIGS. 2E-2J are performed to form the source/drain structures **SD**, remove the portions **224** of the oxide layer **220'**, the upper portions of the silicon rich layers **190** and **410**, the protection layer **210**, and the oxide layer **420**, form the oxide layer **229**, the etch stop layer **230**, the dielectric layer **240**, the gate stacks **G3** and **G4**, the silicide layer **280**, and the contact structures **290**, in accordance with some embodiments.

(157) Since the oxidation of the exposed upper surface **411** of the silicon rich layer **410** occurs after the removal process for removing the upper portions of the silicon rich layer **190**, the oxide layer **420** extends onto the upper surface **411** of the silicon rich layer **410** after the removal process, in accordance with some embodiments. In this step, a semiconductor device structure **400** is substantially formed, in accordance with some embodiments.

(158) Processes and materials for forming the semiconductor device structures **300** and **400** may be similar to, or the same as, those for forming the semiconductor device structure **200** described above. Elements designated by the same reference numbers as those in FIGS. 1A to 4D-2 have the structures, the materials, and the forming methods that are similar or the same. Therefore, the detailed descriptions thereof will not be repeated herein.

(159) FIG. 5 is a flow chart of an example method for fabricating an example semiconductor device structure, in accordance with some embodiments. Referring now to FIG. 5, the flow chart of the method **500** of forming a semiconductor device structure is illustrated according to various aspects of the present disclosure.

(160) The method **500** is merely an example and is not intended to limit the present disclosure beyond what is explicitly recited in the claims. Additional operations can be provided before, during, and after the method **500**, and some operations described can be replaced, eliminated, or moved around for additional embodiments of the method **500**.

(161) At operation **501**, referring to FIG. 5, the method **500** forms a gate stack over a substrate, in accordance with some embodiments. At operation **502**, referring to FIG. 5, the method **500** forms a dielectric layer over the gate stack and the substrate, in accordance with some embodiments.

(162) At operation **503**, referring to FIG. 5, the method **500** forms a first silicon rich layer over the dielectric layer, in accordance with some embodiments. A first atomic percentage of silicon in the first silicon rich layer is greater than about 50%, in accordance with some embodiments. At operation **504**, referring to FIG. 5, the method **500** forms a protection layer over the first silicon

rich layer, in accordance with some embodiments.

(163) At operation **505**, referring to FIG. 5, the method **500** removes a first upper portion and a first bottom portion of the protection layer, in accordance with some embodiments. The protection layer exposes a second upper portion and a second bottom portion of the first silicon rich layer after removing the first upper portion and the first bottom portion, in accordance with some embodiments.

(164) At operation **506**, referring to FIG. 5, the method **500** removes the second upper portion and the second bottom portion of the first silicon rich layer and a third upper portion and a third bottom portion of the dielectric layer under the second upper portion and the second bottom portion, in accordance with some embodiments. The remaining portions of the protection layer, the first silicon rich layer, and the dielectric layer together form a spacer structure, in accordance with some embodiments.

(165) The unique fabrication process flow and the resulting IC device structure of the present disclosure offer advantages over conventional devices. It is understood, however, that no particular advantage is required, other embodiments may offer different advantages, and that not all advantages are necessarily disclosed herein. One advantage is that the yield of the etching process for forming a spacer structure is improved. Specifically, the methods (for forming the semiconductor device structure) form a silicon rich layer in a spacer structure. The silicon rich layer is formed between a dielectric layer thereunder and a protection layer thereover. The silicon rich layer is used as an etch stop layer to prevent over etching of the protection layer thereover and therefore protect the dielectric layer when patterning the protection layer using the etching process. Therefore, the silicon rich layer improves the yield of the etching process.

(166) In accordance with some embodiments, a semiconductor device structure is provided. The semiconductor device structure includes a substrate. The semiconductor device structure includes a gate stack disposed over the substrate. The semiconductor device structure includes a spacer structure disposed over a sidewall of the gate stack. The spacer structure includes a dielectric layer, a silicon rich layer, and a protection layer, the dielectric layer is formed between the gate stack and the silicon rich layer, the silicon rich layer is formed between the dielectric layer and the protection layer, and a first atomic percentage of silicon in the silicon rich layer is greater than about 50%. The semiconductor device structure includes a source/drain structure disposed over the substrate. The spacer structure is formed between the source/drain structure and the gate stack.

(167) In accordance with some embodiments, a semiconductor device structure is provided. The semiconductor device structure includes a substrate. The semiconductor device structure includes a gate stack disposed over the substrate. The semiconductor device structure includes a source/drain structure disposed over the substrate. The semiconductor device structure includes a spacer structure disposed over a sidewall of the source/drain structure. The spacer structure includes a dielectric layer, a first silicon rich layer, and a protection layer, the dielectric layer covers the sidewall of the source/drain structure, the first silicon rich layer is embedded in the dielectric layer, the protection layer is embedded in the first silicon rich layer, and a first atomic percentage of silicon in the first silicon rich layer is greater than about 50%.

(168) In accordance with some embodiments, a method for forming a semiconductor device structure is provided. The method includes forming a gate stack over a substrate. The method includes forming a dielectric layer over the gate stack and the substrate. The method includes forming a first silicon rich layer over the dielectric layer, wherein a first atomic percentage of silicon in the first silicon rich layer is greater than about 50%. The method includes forming a protection layer over the first silicon rich layer. The method includes removing a first upper portion and a first bottom portion of the protection layer. The protection layer exposes a second upper portion and a second bottom portion of the first silicon rich layer after removing the first upper portion and the first bottom portion. The method includes removing the second upper portion and the second bottom portion of the first silicon rich layer and a third upper portion and a third bottom

portion of the dielectric layer under the second upper portion and the second bottom portion, wherein remaining portions of the protection layer, the first silicon rich layer, and the dielectric layer together form a spacer structure.

(169) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A semiconductor device structure, comprising: a substrate; a gate stack disposed over the substrate; a spacer structure disposed over a sidewall of the gate stack, wherein the spacer structure comprises a dielectric layer, a silicon rich layer, and a protection layer, wherein the dielectric layer is disposed between the gate stack and the silicon rich layer, wherein the silicon rich layer is disposed between the dielectric layer and the protection layer, wherein a first atomic percentage of silicon in the silicon rich layer is greater than about 50%; and a source/drain structure disposed over the substrate, wherein the spacer structure is disposed between the source/drain structure and the gate stack, and wherein a portion of a side surface of the source/drain structure extends to a side surface of at least one of: the dielectric layer, the silicon rich layer, or the protection layer.
2. The semiconductor device structure as claimed in claim 1, wherein the first atomic percentage of silicon in the silicon rich layer is greater than a second atomic percentage of silicon in the protection layer.
3. The semiconductor device structure as claimed in claim 1, wherein the spacer structure further comprises an oxide layer disposed over a sidewall of the silicon rich layer.
4. The semiconductor device structure as claimed in claim 3, wherein the oxide layer is in direct contact with the silicon rich layer.
5. The semiconductor device structure as claimed in claim 4, wherein the oxide layer is disposed between the dielectric layer and the protection layer.
6. The semiconductor device structure as claimed in claim 4, wherein the oxide layer is in direct contact with the source/drain structure.
7. The semiconductor device structure as claimed in claim 1, wherein the first atomic percentage of silicon in the silicon rich layer decreases toward the protection layer.
8. The semiconductor device structure as claimed in claim 1, wherein the silicon rich layer is thinner than both the protection layer and the dielectric layer in a direction perpendicular to a sidewall of the gate stack.
9. The semiconductor device structure as claimed in claim 1, further comprising: an oxide layer disposed over an upper surface of the silicon rich layer.
10. A semiconductor device structure, comprising: a substrate; a gate stack disposed over the substrate; a source region and a drain region disposed on opposite sides of the gate stack; and a spacer structure disposed over a sidewall of the source region or the drain region, wherein the spacer structure comprises a dielectric layer, a first silicon rich layer, and a protection layer, wherein the dielectric layer covers the sidewall of the source region or the drain region, wherein the first silicon rich layer is embedded in the dielectric layer, wherein the protection layer is embedded in the first silicon rich layer, and wherein a first atomic percentage of silicon in the first silicon rich layer is greater than about 50%.
11. The semiconductor device structure as claimed in claim 10, further comprising: an oxide layer

disposed over an upper surface of the first silicon rich layer.

12. The semiconductor device structure as claimed in claim 10, wherein the protection layer has a curved upper surface.

13. The semiconductor device structure as claimed in claim 10, further comprising: a second silicon rich layer embedded in the protection layer, wherein a second atomic percentage of silicon in the second silicon rich layer is greater than about 50%.

14. The semiconductor device structure as claimed in claim 13, further comprising: an oxide layer disposed over an upper surface of the second silicon rich layer.

15. A semiconductor device structure, comprising: a substrate; a gate structure disposed over the substrate in a cross-sectional side view; a spacer structure disposed over a sidewall of the gate structure in the cross-sectional side view, wherein the spacer structure comprises a dielectric layer, a silicon-containing layer, and a protective layer, wherein the dielectric layer is disposed between the gate structure and the silicon-containing layer, wherein the silicon-containing layer is disposed between the dielectric layer and the protective layer, wherein the an atomic percentage of silicon in the silicon-containing layer is greater than an atomic percentage of silicon in the protective layer, and wherein an uppermost surface of the protection layer has a lower vertical elevation than an uppermost surface of the dielectric layer in a cross-sectional side view; and a source/drain structure disposed over the substrate in the cross-sectional side view, wherein the spacer structure is disposed between the source/drain structure and the gate structure.

16. The semiconductor device structure of claim 15, wherein the atomic percentage of silicon in the silicon-containing layer is greater than about 50%.

17. The semiconductor device structure of claim 15, wherein the atomic percentage of silicon in the silicon-containing layer decreases toward the protective layer.

18. The semiconductor device structure of claim 15, wherein: the spacer structure further comprises an oxide layer disposed over a sidewall of the silicon-containing layer; and the oxide layer extends to the silicon-containing layer.

19. The semiconductor device structure of claim 18, wherein: the oxide layer is disposed between the dielectric layer and the protective layer; and the oxide layer extends to the source/drain structure.

20. The semiconductor device structure of claim 15, wherein the silicon-containing layer is thinner than the protective layer or the dielectric layer.
